

Database Test

Q1.9

Which of the following refer to data?

Address

Exchange rate

Metallica's last album

An order in an online store

A biography of Alexander Pushkin

Phone number

Process

Insight

Q1.12

Which of the following statements about DBMSs is **INCORRECT**?

DBMSs do not isolate the operations of users working with a DB at the same time.

A DBMS allows you to control user access to data.

A DBMS allows you to manipulate data: insert, update, delete, and select.

A DBMS can be used to protect data from accidents.

Q1.14

In which of the following DB objects is information stored?

Table

Constraint

Data type

Q1.15

Which of the following statements about the primary key is correct?

The primary key is always restricted by uniqueness.

The primary key can contain null values.

Primary key values can contain duplicates.

Q3.13

Which of the following are disadvantages of SQL?

Complexity

Unpredictable performance of complex queries

Requires procedural extensions

DBMS-independent

Standardized

Declarative

Q3.12

What is ANSI SQL?

An SQL standard that defines the syntax of SQL commands.

A structured query language for managing data in relational database management systems

A dialect of SQL that is used in the Microsoft SQL Server RDMS

An artificial, normalized, standardized implementation of SQL.

Q2.13

Which of the following statements is INCORRECT?

For OLAP databases, high-level normalization is not important.

The higher the level of normalization, the more relationships there are, and the more connections there are between them.

Data warehouses are optimized to ensure high performance when selecting a large amount of data.

For OLTP databases, it is not important to insert and update data.

Q4.27

Which of the following statements about the TRUNCATE statement is true?

It removes ALL of the rows from a table.

It is not MVCC-safe; the effect is visible to other transactions instantly.

The WHERE clause cannot be used.

It removes specific rows from a table.

It inserts rows in a table.

It deletes a table's structure permanently.

Q4.16

Which of the following is the correct way to insert multiple rows in a single INSERT statement?

INSERT INTO table_name (column1, column2) VALUES (value1, value2), (value3, value4), (value5, value6)

INSERT INTO table_name (column1, column2) (value1, value2), (value3, value4), (value5, value6)

INSERT INTO table_name (value1, value2), (value3, value4), (value5, value6) VALUES (column1, column2)

INSERT INTO table_name VALUES (column1, column2) (value1, value2), (value3, value4), (value5, value6)

Q4.15

Which of the following statements is correct?

DML and DDL commands are transaction-safe and can be rolled back if necessary.

Database-wide and tablespace-wide operations (including DCL operators) can be rolled back

Any non-committed transaction will be automatically committed if the session is interrupted.

All DML commands except TRUNCATE are MVCC-unsafe.

Q4.18

Which of the following is the correct way to insert data into a table using a subquery?

INSERT INTO table_name (column1, column2) SELECT column1, column2 FROM other_table WHERE condition

INSERT INTO table_name (column1, column2) VALUES (SELECT column1, columns2 FROM other_table WHERE condition)

INSERT INTO table_name SELECT column1, column2 FROM other_table WHERE condition

INSERT SELECT column1, column2 INTO table_name FROM other_table WHERE condition

Q5.29

What is the purpose of the GENERATED ALWAYS AS statement in PostgreSQL?

It is used to generate a computed column based on a specified expression.

It is used to create a new column in a table.

It is used to modify the data type of an existing column in a table

It is used to drop a column from a table

Q5.27

Which of the following is the correct syntax for adding a PRIMARY KEY constraint to a table column?

ADD CONSTRAINT PRIMARY KEY ([column_name]);

ADD PRIMARY KEY ([column_name]);

ADD [column_name] PRIMARY KEY;

ADD PRIMARY KEY TO [column_name];

Q5.31

What is the common table expression (CTE) in PostgreSQL?

A named temporary set of results that can be used in a SELECT, INSERT, UPDATE, or DELETE statement

A temporary table that is created to hold intermediate results

A virtual table that is defined by a SELECT statement

A backup of a table's contents that can be restored in case of data loss.

Q0.0

Which of the following statements about views are true?

Views can be queried like regular tables but cannot be modified directly.

Views can be used to enforce security and access controls.

Views can be used in the SELECT statement.

Views can contain only one SELECT query.

A view must return a value after execution.

Views can accept input parameters, perform calculations, and return output values.

Views can contain several SELECT queries.

Views allow exceptions and errors to be handled.

Views can handle more complex logic using control structures like IF-ELSE and FOR loops.

A view can be parsed line by line.

Q5.32

How is a table snapshot different from a view?

A view is always up to date with the underlying table, while a table snapshot is a point-in-time copy.

A view is a read-only copy of a table, while a table snapshot is read-write.

A view is created by executing a SELECT statement, while a table snapshot is created using a special command.

A view can be used to update the underlying table, while a table snapshot cannot.

Q6.18

In what situations are procedural language functions more appropriate than query language functions or views?

When the logic needs to be encapsulated and modularized

When the logic requires complex business logic and calculations

When the logic requires parameterization

When the logic needs to be optimized for performance

When the logic requires exceptions and errors to be handled

When providing a way to control access to sensitive data

When the logic requires simple joins and filtering

Q6.13

What are INOUT parameters in a user-defined function in PostgreSQL?

Parameters that can be both passed into the function from outside and used to return values to the calling program

Parameters that are passed into the function from outside

Parameters that are declared inside the function and can be used to pass values to other functions

Parameters that are declared inside the function and can be used to return values to the calling program

Q6.12

Which procedural language can be used to write user-defined functions in PostgreSQL?

Query language functions (function written in SQL)

Procedural language functions (functions written in PL/pgSQL)

Internal functions

C-language functions

Q7.11

What is the purpose of auditing in PostgreSQL?

To detect and track security breaches

To authenticate users

To encrypt communication between client and server

To back up a database

Q8.9

What is an OLTP system?

A system that facilitates and manages transaction-oriented applications, typically for the purpose of processing data entry and retrieval transactions

A system that facilitates and manages transaction-oriented applications, typically for the purpose of reporting with heavy analytical processing

A system that facilitates and manages any database applications, typically for ad hoc analysis

A system that facilitates and manages cloud applications, typically for the purpose of reporting analytical data in separate transactions

Q7.12

What is the purpose of the EXECUTE privilege in PostgreSQL?

To grant the ability to execute stored procedures or functions

To grant the ability to modify a table

To grant the ability to view data in a table

To grant the ability to create a table

Q8.12

Which of the following are the goals of DWH?

To present an organization's information consistently

To be adaptive and resilient to change

To be accepted by the business community

To serve as operational data storage for OLTP systems.

Q1.16

What is the purpose of a foreign key in a DB?

GPT version:

A **foreign key** ensures referential integrity by linking a column in one table to the primary key in another table, enforcing valid relationships between data in the tables.

My version:

The purpose of a foreign key is to connect two tables and establish a relationship between them. The connection will be by linking a column in one table to the primary key in another table.

Q1.13

Why are constraints used in DBs?

GPT version:

Constraints are used in databases to enforce **data integrity** and ensure the accuracy, validity, and consistency of the data. They define rules that the data must follow and prevent invalid data from being entered into the database.

My version:

Constraints are used in DBs to enforce data integrity and ensure validation of the data. They ensure that data in the database is valid, and the invalid input data cannot be entered into the databases

Q2.10

Describe 1st normal form.

GPT version:

First Normal Form (1NF) ensures that a table's data is organized by requiring each column to have atomic (indivisible) values, no repeating groups or arrays, and unique rows identifiable by a primary key.

My version:

The 1NF means that the table is simple and clean. Each column should have one value and there shouldn't be repeating columns like "Phone1", "Phone2".

Q2.12

Describe 3rd normal form.

GPT version:

Third Normal Form (3NF) ensures that every column in a table depends only on the primary key and not on other non-key columns, eliminating transitive dependencies. It makes data cleaner and avoids redundant relationships.

My version:

3NF means that data in the table should depend only on the primary key and no other data. It helps to avoid storing extra or unnecessary data and connection.

Q3.17

Rewrite the following query using a subquery:

```
SELECT i.name,  
i.description
```

```
FROM Item i  
INNER JOIN Item_sale is ON i.item_id = is.item_id;
```

Solution:

```
SELECT i.name,  
i.description  
FROM Item i  
WHERE i.item_id IN (SELECT is.item_id FROM Item_sale is)
```

Q3.18

Rewrite the following query using CTE:

```
SELECT i.item_id,  
i.name,  
i.description  
FROM Item i  
WHERE i.item_id IN (SELECT is.item_id  
FROM Item_sale is  
INNER JOIN customer c ON is.cust_id = c.cus_id  
WHERE c.cust_name = 'John');
```

Solution:

```
WITH Item_sales_for_John AS (  
SELECT is.item_id  
FROM Item_sale is  
INNER JOIN customer c ON is.cust_id = c.cus_id  
WHERE c.cust_name = 'John')
```

```
SELECT i.item_id, i.name, i.description
FROM Item i
WHERE i.item_id in (SELECT item_id
FROM Item_sales_for_John);
```

Q3.20

Define INTERSECT set operator and list its properties (minimum 100 words).

The **INTERSECT** set operator in SQL is used to return the common rows between the result sets of two or more SELECT statements. It compares the datasets and includes only those rows that are present in all the queries being intersected. This operator is useful for finding overlapping data in different tables or queries.

Q4.17

What is a TCL statement? Write a response of at least 100 words.

A **TCL (Transaction Control Language)** statement in SQL is used to manage and control transactions in a database. Transactions are sequences of operations that are executed as a single logical unit of work. TCL statements help ensure the consistency and integrity of the database by allowing users to commit, rollback, or set savepoints within transactions.

Q4.19

Correct the syntax mistakes by rewriting the query below:

```
INSERT IN table_name (column1_name, column2_name):
ROWS (column1_value, column2_value)
; (column1_value, column2_value)
```

Solution:

```
INSERT INTO table_name (column1_name, column2_name)
VALUES (column1_value, column2_value), (column1_value, column2_value);
```

Q4.21

Write a query for writing a script to insert in the “country” table a row with country_name “South Sudan” and that returns a new value of country_id.

```
CREATE TABLE country (  
country_id SERIAL NOT NULL,  
country_name TEXT NOT NULL,  
last_update TIMESTAMPTZ NOT NULL DEFAULT now(),  
CONSTRAINT country_pkey PRIMARY KEY (country_id));
```

Solution:

```
INSERT INTO country (country_name)  
VALUES ('South Sudan')  
RETURNING country_id;
```

Q4.28

Correct the syntax mistakes by rewriting the query below:

```
TRUNCATE FROM table_name  
WHERE condition (write with two options);
```

Solution:

1-option:

```
DELETE FROM table_name  
WHERE condition;
```

2-option:

```
TRUNCATE table_name
```

Q4.30

Describe the differences between **DELETE** and **TRUNCATE** statements. Use the following criteria: effect, performance, server impact, table lock, transactional, MVCC-safe. Write a response of at least 100 words.

The **DELETE** and **TRUNCATE** statements are used to remove rows from a table, but they differ significantly in their behavior and use cases.

1. Effect:

- a. **DELETE** removes specific rows based on a **WHERE** condition or all rows without it.
- b. **TRUNCATE** removes all rows from a table without conditions, resetting it.

2. Performance:

- a. **DELETE** is slower as it processes each row individually.
- b. **TRUNCATE** is faster because it bypasses row-by-row deletion and deallocates table data in bulk.

3. Server Impact:

- a. **DELETE** generates more logs, increasing I/O load.
- b. **TRUNCATE** generates minimal logs, reducing server impact.

4. Table Lock:

- a. **DELETE** uses row-level locks.
- b. **TRUNCATE** uses a table-level lock.

5. Transactional:

- a. **DELETE** is transactional and can be rolled back.
- b. **TRUNCATE** is partially transactional and cannot be rolled back in some databases.

6. MVCC-Safe:

- a. **DELETE** is MVCC-safe, maintaining multi-version concurrency.
- b. **TRUNCATE** is not MVCC-safe and its effects are immediate.

Use **DELETE** for precise row removal and **TRUNCATE** for bulk, fast deletion.

Q4.31

Remove all the data from the “country” table in the fastest way possible.

TRUNCATE TABLE country;

Q5.30

Write script to create the table "Customer":

Add the following into the code for creating the table:

- Primary key constraint to Customer_ID column;
- Foreign key constraint to RegionCustomer_ID column, that refers to RegionCustomer_ID column of RegionCustomer table;
- Default constraint with date of inserting CreatedDate column

Customer table:

CUSTOMER		
COLUMN	DATATYPE	NULL / NOT NULL
Customer_ID	BIGSERIAL	NOT NULL
CustomerNumber	TEXT	NOT NULL
NumberBill	TEXT	NULL
CustomerLegalName	TEXT	NOT NULL
RegionCustomer_ID	INTEGER	NOT NULL
ISActive	BOOLEAN	NOT NULL
ClassificationCust_ID	INTEGER	NULL
CreatedDate	DATE	NOT NULL
CreatedBy	TEXT	NOT NULL

Solution:

```
CREATE TABLE Customer (
```

```
Customer_ID BIGSERIAL PRIMARY KEY,
```

```
CustomerNumber TEXT NOT NULL,
```

```
NumberBill TEXT,
```

```
CustomerLegalName TEXT NOT NULL,
```

```
RegionCustomer_ID INTEGER NOT NULL,
```

```
ISActive BOOLEAN NOT NULL,
```

```
ClassificationCust_ID INTEGER,
```

```
    CreatedDate DATE NOT NULL DEFAULT CURRENT_DATE,  
    CreatedBy TEXT NOT NULL,  
    CONSTRAINT FK_RegionCustomer FOREIGN KEY (RegionCustomer_ID)  
    REFERENCES RegionCustomer (RegionCustomer_ID)  
)
```

Q5.36

Write a query for writing a script to remove the CreatedBy column from the “Customer” table.

```
ALTER TABLE Customer  
DROP COLUMN CreatedBy;
```

Q5.37

Write a query for writing a script to change the name of the CreatedDate column in the “Customer” table to LastUpdatedDate.

```
ALTER TABLE Customer  
RENAME COLUMN CreatedDate TO LastUpdatedDate;
```

Q5.38

Write a query for writing a script to change the data type of the ClassificationCustID column in the “Customer” table to BIGINT.

```
ALTER TABLE Customer  
ALTER COLUMN ClassificationCust_ID TYPE BIGINT;
```

Q5.39

Write a query for writing a script to add a Default constraint with the value TRUE to the IActive column in the “Customer” table.

```
ALTER TABLE Customer
```

```
ALTER COLUMN IActive SET DEFAULT TRUE;
```

Q5.40

Write a query for writing a script to remove the NOT NULL constraint on the CustomerLegalName column from the “Customer” table.

```
ALTER TABLE Customer
```

```
ALTER COLUMN CustomerLegalName DROP NOT NULL;
```

Q8.13

Which of the following best describes the structure of the star schema?

It uses a complex network of interconnected tables.

It employs a highly normalized structure with many levels of relationships.

It consists of a central fact table connected to multiple dimension tables.

It is designed for distributed data store across multiple nodes.

Q9.8

The RANK function can be made to operate within groups (i.e., the RANK gets reset whenever the group changes). What clause can this be accomplished with?

HAVING

PARTITION BY

ORDER BY

ROWS / RANGE

Q9.7

Which of the following statements are correct?

DENSE_RANK leaves gaps in the ranking sequence if there is a tie between the ranks of the preceding records.

RANK assigns a unique number (starting from 1, as defined by the order) to each row within a partition.

ROW_NUMBER assigns a unique number (starting from 1, as defined by the order) to each row within a partition.

DENSE_RANK does not skip any ranks if there is a tie between the ranks of the preceding records.

Q9.10

Which function divides an ordered partition into a specified number of groups called buckets and assigns a bucket number to each row in the partition?

NTH_VALUE

PERCENT_RANK

NBUCKET

RATIO_TO_REPORT

NTILE

Q9.9

Complete the following statement: The result of the CUME_DIST (X) function is calculated as ...

... the number of values before and including X in the specified order/number of records in the current group.

Q10.6

What is the purpose of a window frame in the context of window function?

It determines the order of rows within a window.

It defines the size of the result set.

It defines the subset of rows used for calculations within a window.

It specifies how rows are partitioned within a window.

Q10.8

Which of the following statements about window frames are correct?

frame_start can be FOLLOWING.

frame_end choice can appear earlier than the frame_start choice (for example, RANGE BETWEEN CURRENT ROW AND offset PRECEDING).

frame_start cannot be UNBOUNDED FOLLOWING.

frame_end choice cannot appear earlier than the frame_start choice (for example, RANGE BETWEEN CURRENT ROW AND offset PRECEDING).

frame_end can be UNBOUNDED PRECEDING.

frame_end cannot be UNBOUNDED PRECEDING

Q10.11

What is the default option for the clause FRAME_EXCLUSION?

EXCLUDE TIRES

EXCLUDE CURRENT ROW

EXCLUDE GROUP

EXCLUDE NO OTHERS

Q10.9

A FRAME_END of the current row means the frame ends with the current row's last peer row ...

In RANGE mode.

In ROWS mode.

In GROUPS mode.

Q10.10

How many rows will be included in the frame RANGE BETWEEN interval '1' day PRECEDING AND Interval '3' day FOLLOWING?

How many rows will be included in the frame
RANGE BETWEEN interval '1' day PRECEDING AND interval '3' day FOLLOWING?

01-Jan-2020
02-Jan-2020
04-Jan-2020
05-Jan-2020
07-Jan-2020
07-Jan-2020 current row
08-Jan-2020
09-Jan-2020
09-Jan-2020
23-Jan-2020
24-Jan-2020

4

Q9.11

What is the purpose of the Lag function in SQL?

To retrieve the value of the next row within the partition.

To retrieve the last value within the partition.

To retrieve the first value within the partition.

To retrieve the value of the previous row within the partition.

Q7.13

What is a group role in PostgreSQL?

A role that has administrative privileges.

A role that has the ability to create new roles.

A role that can be used to log in to a database.

A role that can be used to group other roles together.

Q5.42

Can you drop a table with dependencies using the SQL DROP TABLE CASCADE command if you want to keep the dependent objects?

Yes, the ALTER TABLE command must be used to remove the dependencies first.

No, the dependent objects will always be deleted along with the table.

Q5.25

Which SQL DDL statement is used to delete a table?

TRUNCATE TABLE

ALTER TABLE

DELETE TABLE

DROP TABLE

Q5.43

Why is it necessary to minimize the number of users that can drop tables in a production environment? (Write a response of at least 100 words).

Minimizing the number of users who can drop tables in a production environment is essential to maintain the integrity, stability, and security of the database. Dropping a table is a destructive operation that permanently deletes both the data and the table structure, potentially leading to critical data loss if done accidentally or maliciously. In a production environment, such actions can disrupt business operations, cause downtime, and result in significant financial or reputational damage. By restricting this privilege to a limited number of trusted and experienced users, organizations can reduce the risk of errors or abuse. Proper auditing and role-based access control further ensure accountability and help safeguard critical data assets.

Q5.42

Write a query for writing a script to drop the Customers2019Snapshot table.

DROP TABLE Customers2019Snapshot;

Q5.41

Write a query for writing a script to add a CHECK constraint to the MarketNumber column with the following rule: The value in the MarketNumber column must be greater than 0;

ALTER TABLE your_table_name

ADD CONSTRAINT chk_marketnumber_positive

CHECK (MarketNumber > 0);

Q0.0

Write a query for writing a script to create a view called ActiveCustomers that contains only active customers. The view should include the columns Customer_ID, CustomerNumber, NumberBill, CustomerLegalName, and RegionCustomer_ID.

CREATE VIEW ActiveCustomers AS

SELECT

```
Customer_ID,  
CustomerNumber,  
NumberBill,  
CustomerLegalName,  
RegionCustomer_ID  
FROM  
Customers  
WHERE  
IsActive = 1;
```

Q5.33

Write a query for writing a script to create a snapshot table called Customer2019Snapshot of the customer data that was received in 2019.

The snapshot should include the columns Customer_ID, CustomerNumber, NumberBill, and CustomerLegalName.

```
CREATE TABLE Customer2019Snapshot AS  
SELECT  
Customer_ID,  
CustomerNumber,  
NumberBill,  
CustomerLegalName  
FROM Customer  
WHERE EXTRACT(YEAR FROM ReceiveDate) = 2019;
```

Q2.9

For which of the following purposes do databases need to be normalized?

To prevent data anomalies when manipulating data and ensure data integrity.

To increase data redundancy in a table, which simplifies data analysis processes.

To ensure the values in a table column are unique.

To control user access to data.

Q1.18

Which of the following statements about relations in relational database theory is INCORRECT?

The values of tuples and attributes must be unique.

Attribute values must be atomic.

Attribute values are equal to tuple values.
Tuple and attribute values are unordered.